FLOW MATCHING AND OPTIMAL TRANSPORT: AN ILLUSTRATED INTRODUCTION TO WASSERSTEIN–OTTO GEOMETRY

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Abstract. We recall and derive the equations underlying Flow Matching, making explicit the probability density function of the linearly interpolated random variable and relating it to the transporting velocity field through the continuity equation. We visualise how the initial density is transported and compare the resulting path with a geodesic in Wasserstein–Otto geometry. We demonstrate that, for an isotropic Gaussian initial law, the Flow Matching velocity field is the minimum-energy representative of the density variation at each time, among fields satisfying the continuity equation, even though the path is generally not a geodesic under the Wasserstein–Otto metric. We explain the distinction between this local minimisation and the minimisation of the action over paths with fixed endpoints using explicit calculations and illustrations.

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1. Introduction

Generative machine learning consists in approximating a probability density p_{data}, from independent and identically distributed samples $x_1, \dots, x_N \sim p_{\text{data}}$, using a model p_θ. A classical approach is to estimate an energy-based parametric model

$$p_\theta(x) = Z_\theta^{-1} e^{-U_\theta(x)}, \qquad Z_\theta = \int_E e^{-U_\theta(x)}\, dx,$$

whose parameters are optimised by minimising the Kullback–Leibler divergence $D_{\text{KL}}(p_{\text{data}} \| p_\theta)$, or another discrepancy. This approach and the difficulties arising from it are discussed in Mallat's lectures at the Collège de France [18]. Computing the normalising constant Z_θ and the model expectations appearing in likelihood-based training involves integrals which are generally intractable in high dimension, as for images or videos. Even after training, samples must be drawn from the model. For energy-based models, this is often done with Markov chain Monte Carlo methods, such as the Metropolis algorithm [19].

A recent approach combines distribution estimation and sampling by learning how to transport a simple distribution p_{init} to the data distribution. Diffusion models [20, 21] progressively add noise to data until the resulting distribution is close to a prescribed noise law. A learned reverse process then generates data by progressively removing this noise. Flow Matching [2] instead learns a velocity field associated with a chosen probability path. In the case studied here, this path results from linear interpolation between a random variable drawn from the data distribution and an independent random variable drawn from the initial law. Sampling is performed by integrating this field from the initial distribution.

This manuscript takes a pedagogical point of view on the probability path and velocity field arising from linear interpolation in Flow Matching. We derive the density of the interpolated random variable, relate its evolution to the transporting velocity through the continuity equation, and illustrate the resulting flow. We then use Wasserstein–Otto geometry to address two questions. At each time, is the Flow Matching velocity a minimum-energy field transporting the prescribed density variation? Does the density path itself follow a geodesic between its endpoints? Separating these questions helps explain the role of the velocity field and of the chosen probability path in Flow Matching. We discuss them through explicit calculations and a Gaussian mixture example.

2. Flow Matching and transport of densities

Flow Matching belongs to the family of methods which estimate the probability distribution underlying the available data by transporting an initial distribution p_{init}. The idea is to approximate p_{data} by the push-forward of p_{init} under a flow ϕ_t associated with a velocity field $v_\theta(t, x)$. For each initial

point x, its trajectory $t \mapsto \phi_t(x)$ satisfies

$$\partial_t \phi_t(x) = v_\theta(t, \phi_t(x)), \qquad \phi_0(x) = x.$$

The flow transports the initial distribution, and its value at $t = 1$ defines the model distribution:

$$p_{\text{data}}\, dy \simeq (\phi_1)_\#(p_{\text{init}}\, dx).$$

The velocity field v_θ is to be learned. Flow Matching represents it by a neural network and learns it by regression, without differentiating through the numerical solution of the flow equation [2]. Related formulations include rectified flow [7]. They have been used for high-resolution image synthesis [8], smaller one-step image generators [9], and image editing [10]. Applications also include speech generation with Voicebox [11], speech and sound generation with Audiobox [12], video and audio generation with Movie Gen [13], and audio-driven portrait animation with FLOAT [14].

2.1. **Training and sampling**

Training. We use the data-to-initial time convention for training, and reverse it when sampling. A neural network $v_\theta(t, y)$ represents the velocity in this convention. Flow Matching allows several choices of conditional probability paths; here we use independent $X \sim p_{\text{data}}$, $Z \sim p_{\text{init}}$ and the linear interpolation

$$X_t = (1 - t)X + tZ, \qquad \dot{X}_t = Z - X.$$

With t sampled uniformly on $[0, 1]$, independently of (X, Z), the conditional regression loss is

$$\mathcal{L}(\theta) = \mathbb{E}_{t,X,Z}\left[\|v_\theta(t, (1 - t)X + tZ) - (Z - X)\|^2\right]. \tag{1}$$

Training is relatively simple: evaluating the expectation requires sampling Z from p_{init}, drawing X from the available data, and sampling t uniformly in $[0, 1]$. These draws are independent. The loss is estimated on minibatches and θ is updated by stochastic gradient methods. The target $Z - X$ is directly available, so no ODE simulation is needed during training. The linear interpolation is one choice within the conditional Flow Matching framework [2].

At the population level, minimisation over square-integrable fields gives

$$v_t(y) = \mathbb{E}[Z - X \mid X_t = y]. \tag{2}$$

This follows from the conditional-mean characterisation of least-squares regression.

Sampling. We draw $Y_0 \sim p_{\text{init}}$ and integrate the learned velocity in reverse time:

$$\frac{dY_s}{ds} = -v_\theta(1 - s, Y_s), \qquad 0 \le s \le 1.$$

The generative flow ϕ_s introduced above therefore uses the velocity $-v_\theta(1 - s, \cdot)$, and its endpoint gives the model distribution $(\phi_1)_\#(p_{\text{init}}\, dx)$. In practice, the integration is performed with a numerical solver, such as explicit Euler or a Runge–Kutta method.

2.2. Velocity fields and density evolution

Consider a smooth velocity field v_t and its flow ϕ_t, with $\partial_t \phi_t = v_t \circ \phi_t$ and $\phi_0 = \mathrm{Id}$. Assume that this flow is a diffeomorphism and that p_t is a smooth transported density. For a bounded domain Ω with smooth boundary, define the moving domain

$$\Omega_t = \phi_t(\Omega).$$

Preservation of mass gives

$$\int_{\Omega_t} p_t(y)\, dy = \int_{\Omega} p_0(x)\, dx.$$

Differentiating with Reynolds' transport theorem yields

$$0 = \frac{d}{dt} \int_{\Omega_t} p_t(y)\, dy = \int_{\Omega_t} (\partial_t p_t + \mathrm{div}(p_t v_t))(y)\, dy.$$

Since this holds for every such domain, we obtain the continuity equation

$$\partial_t p_t + \mathrm{div}(p_t v_t) = 0.$$

This equation links the temporal derivative of the density p_t to the velocity field transporting it. The Wasserstein–Otto metric will rely on this relation to measure the kinetic cost of a density variation; we return to it in Section 4.

Figure 1 illustrates this preservation of mass in a domain moving with the flow.

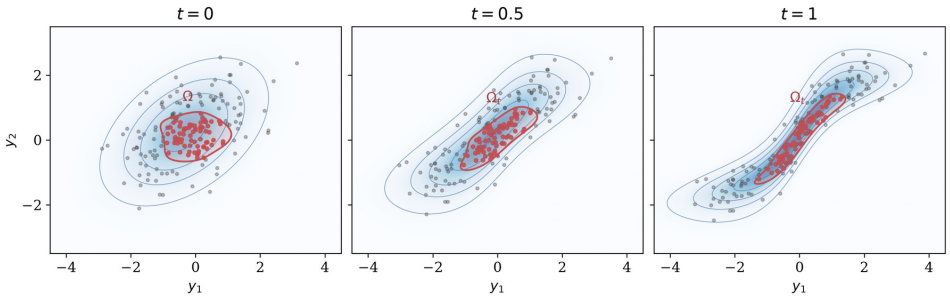


FIGURE 1. A density and a material domain transported by a smooth flow. The red points remain in $\Omega_t = \phi_t(\Omega)$, whose probability mass is constant. Reynolds' transport theorem gives $\partial_t p_t + \mathrm{div}(p_t v_t) = 0$. Adapted from [23].

3. Probability paths under independent interpolation

We consider a Euclidean space $E \simeq \mathbb{R}^d$, with its Lebesgue measure, and two densities $p_{\mathrm{data}}, p_{\mathrm{init}} \in \mathcal{P}_+(E)$ with finite second moments. Let X and Z be independent random variables with densities p_{data} and p_{init}, respectively. We study the interpolation

$$X_t = (1-t)X + tZ, \qquad 0 \le t \le 1. \tag{3}$$

Here time runs from the data distribution to the initial distribution: $p_0 = p_{\text{data}}$ and $p_1 = p_{\text{init}}$. The generative direction is obtained by reversing time. Independence is part of the construction; a different coupling of the same endpoints may give a different density path.

3.1. Contracted densities and convolution

The interpolation map and its fibres. For fixed t, introduce the linear map

$$\psi_t : E \times E \longrightarrow E, \qquad \psi_t(x, z) = (1-t)x + tz.$$

The law of X_t is the push-forward of the product measure $p_{\text{data}}(x)p_{\text{init}}(z)\,dx\,dz$ by ψ_t. In orthonormal coordinates its differential has matrix

$$J_t := D\psi_t = \operatorname{Mat}(d\psi_t) = \begin{pmatrix} (1-t)\operatorname{Id} & t\operatorname{Id} \end{pmatrix}.$$

Thus ψ_t is surjective, and its normal volume factor is

$$A_t := \sqrt{\det(J_t J_t^{\mathsf{T}})} = \left((1-t)^2 + t^2\right)^{d/2}. \tag{4}$$

The density $p_t = p_{X_t}$ is obtained by integrating the joint density of (X, Z) over the fibre $\psi_t^{-1}(y)$. Applying the coarea formula to this push-forward [1], we obtain

$$p_t(y) = \frac{1}{A_t} \int_{\psi_t^{-1}(y)} p_{\text{data}}(x)p_{\text{init}}(z)\,d\mathcal{H}^d(x, z). \tag{5}$$

Here $\mathcal{H}^d$ is the induced Euclidean volume on the fibre, and $A_t > 0$ is the constant normal volume factor computed in (4).

For $0 < t < 1$, parametrise the fibre by

$$F_{t,y}(x) = \left(x, \frac{y - (1-t)x}{t}\right).$$

Writing $DF_{t,y}$ for its Jacobian matrix in orthonormal coordinates, its induced volume element is

$$\sqrt{\det((DF_{t,y})^{\mathsf{T}} DF_{t,y})}\,dx = \left(1 + \frac{(1-t)^2}{t^2}\right)^{d/2} dx = \frac{A_t}{t^d}\,dx.$$

Substitution into (5) cancels A_t and yields

$$p_t(y) = \int_E p_{\text{data}}(x)\,t^{-d} p_{\text{init}}\left(\frac{y - (1-t)x}{t}\right) dx. \tag{6}$$

Convolution form. Rescaling the integration variable in (6) gives

$$\boxed{p_t = \left[\frac{1}{(1-t)^d}\,p_{\text{data}}\left(\frac{\cdot}{1-t}\right)\right] * \left[\frac{1}{t^d}\,p_{\text{init}}\left(\frac{\cdot}{t}\right)\right], \qquad 0 < t < 1.} \tag{7}$$

The initial and data distributions are contracted by t and $1-t$, respectively, and then convolved. When t is close to zero, p_{data} is almost unchanged, while the contracted initial distribution is close to a Dirac mass at the origin. The convolution is therefore close to p_{data}. When t is close to one, we have the opposite phenomenon: the contracted data distribution is close to a Dirac mass at the origin, while p_{init} is almost unchanged. The convolution is then close to p_{init}. Figure 2 illustrates this evolution.

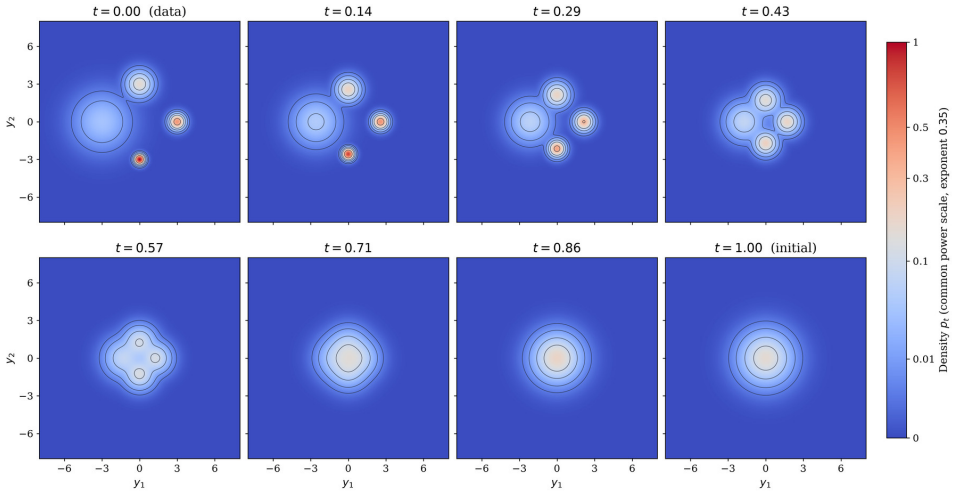


FIGURE 2. Evolution of p_t from a Gaussian mixture p_{data} to the Gaussian $p_{\text{init}} = \mathcal{N}(0, \text{Id})$. Colours and contours represent the density at eight times from $t = 0$ (top left) to $t = 1$ (bottom right).

3.2. The initial distribution and the marginal velocity

For $0 < t < 1$, assume that the endpoint densities are smooth and rapidly decreasing, so that we may differentiate under the integral in (6). Directly,

$$\begin{aligned}\partial_t p_t(y) &= \int_E p_{\text{data}}(x)\, \partial_t \left[\frac{1}{t^d} p_{\text{init}}\left(\frac{y-(1-t)x}{t}\right)\right] dx \\ &= \int_E p_{\text{data}}(x) \left[-\frac{d}{t^{d+1}} p_{\text{init}}\left(\frac{y-(1-t)x}{t}\right)\right. \\ &\qquad \left. + \frac{1}{t^d}\left\langle \nabla p_{\text{init}}\left(\frac{y-(1-t)x}{t}\right), \frac{x-y}{t^2}\right\rangle\right] dx. \quad (8)\end{aligned}$$

We now use the continuity equation derived in Section 2.2,

$$\partial_t p_t + \operatorname{div}(p_t v_t) = 0$$

to find a velocity field transporting this density path. Writing the result of (8) as a divergence gives

$$\partial_t p_t(y) = -\operatorname{div}_y\left(\int_E \frac{y-x}{t}\, p_{\text{data}}(x) \frac{1}{t^d} p_{\text{init}}\left(\frac{y-(1-t)x}{t}\right) dx\right).$$

We may therefore choose the flux

$$p_t(y) v_t(y) = \int_E \frac{y-x}{t}\, p_{\text{data}}(x) \frac{1}{t^d} p_{\text{init}}\left(\frac{y-(1-t)x}{t}\right) dx.$$

Dividing by $p_t(y)$ and using (6), we obtain

$$\boxed{v_t(y) = \frac{\displaystyle\int_E \frac{y-x}{t}\, p_{\text{data}}(x) t^{-d} p_{\text{init}}\left(\frac{y-(1-t)x}{t}\right) dx}{\displaystyle\int_E p_{\text{data}}(x) t^{-d} p_{\text{init}}\left(\frac{y-(1-t)x}{t}\right) dx}.} \tag{9}$$

To interpret this field, first consider the joint law of (X, X_t). The change of variables $(x, z) \mapsto (x, (1-t)x + tz)$ has determinant t^d, so its joint density is

$$p_{X,X_t}(x, y) = \frac{p_{\text{data}}(x)}{t^d}\, p_{\text{init}}\left(\frac{y-(1-t)x}{t}\right).$$

Dividing by the marginal density $p_{X_t}(y) = p_t(y)$ gives

$$p_{X|X_t=y}(x) = \frac{p_{X,X_t}(x, y)}{p_{X_t}(y)} = \frac{p_{\text{data}}(x)}{t^d p_t(y)}\, p_{\text{init}}\left(\frac{y-(1-t)x}{t}\right).$$

On the fibre $(1-t)x + tz = y$, we have $(y-x)/t = z - x$. Hence

$$v_t(y) = \frac{y - \mathbb{E}[X \mid X_t = y]}{t} = \mathbb{E}[Z - X \mid X_t = y]. \tag{10}$$

This is the marginal velocity associated with the conditional straight trajectories, and the population target of the velocity regression [2, 3]. Figure 3 shows this velocity field for the same distributions as in Figure 2.

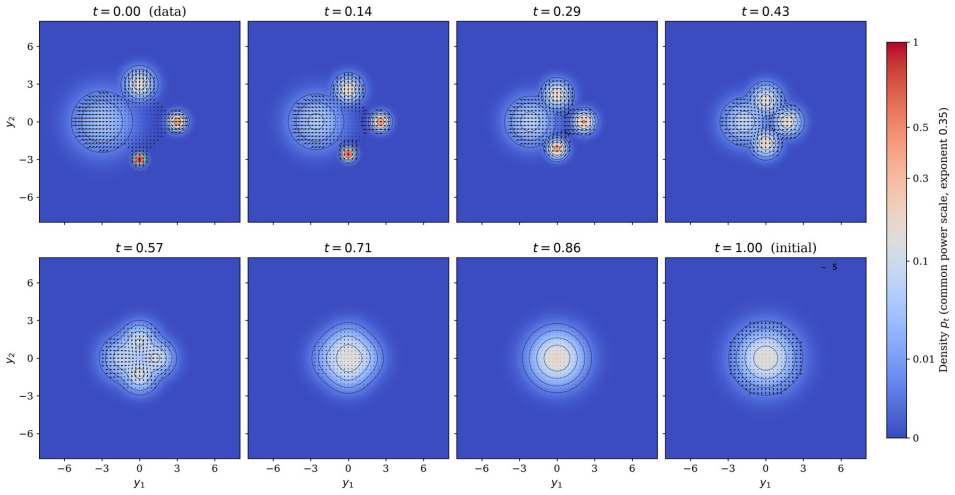


FIGURE 3. The same Gaussian mixture to Gaussian interpolation, with the marginal velocity v_t evaluated on a regular grid of spacing 0.3, shown near the density and between the Gaussian components. Arrows point in the data-to-initial-distribution direction; their lengths use the same scale in all panels.

These formulas describe the data-to-initial-distribution direction. Reversing time gives $\widetilde{p}_s = p_{1-s}$ and $\widetilde{v}_s = -v_{1-s}$. Expressions involving $1/t$ are used only for $0 < t < 1$; endpoints must be treated separately.

To sample in the generative direction, we draw Y_0 from p_{init} and solve

$$\frac{dY_s}{ds} = -v_{1-s}(Y_s), \qquad 0 \le s \le 1.$$

This ODE can be integrated with Euler's method or another numerical solver. Figure 4 shows trajectories computed with a fourth-order Runge–Kutta scheme.

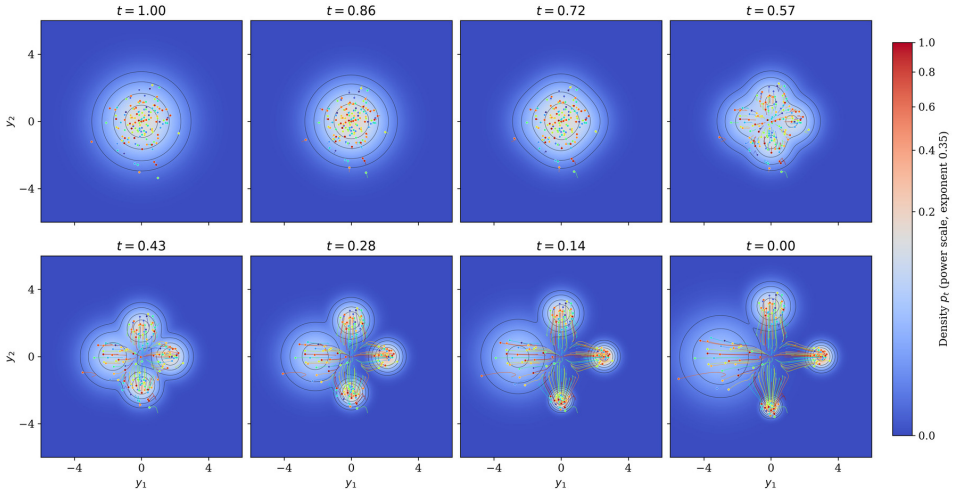


FIGURE 4. Numerical integration of 150 particles from the Gaussian p_{init} to the mixture p_{data}. Time is reversed: $t = 1-s$ decreases from 1 (top left) to 0 (bottom right). Coloured curves retain the past trajectories; dots mark current positions. The background shows the exact density p_t.

4. Wasserstein–Otto geometry and comparison with geodesics

For a broader comparison of Fisher–Rao and Wasserstein–Otto geometries, we refer to [23].

We keep the same endpoints p_{data} and p_{init}, with finite second moments, and the same time direction. We now describe what would be required of a Wasserstein–Otto geodesic between them. Write p_t^{geo} for this reference path, to distinguish it from the independently coupled interpolation studied above.

4.1. Tangent space of probability densities

We briefly recall the density representation of probability measures. On $E \simeq \mathbb{R}^d$, a probability measure P absolutely continuous with respect to Lebesgue measure $\mathcal{L}^d$ admits a Radon–Nikodym density $p = dP/d\mathcal{L}^d$, so that [15, Chapter 3]

$$P(A) = \int_A p(x)\,dx, \qquad \int_E p(x)\,dx = 1$$

for every Borel set $A \subset E$. We work with the space of strictly positive probability densities

$$\mathcal{P}_+(E) = \left\{ p > 0 : \int_E p(x)\,dx = 1 \right\},$$

with finite second moments and the regularity required by the following computations. In the formal smooth description, a vector in the tangent space $T_p\mathcal{P}_+(E)$ is a density variation u with $\int_E u\,dx = 0$; along a differentiable probability path, $u = \partial_t p_t$ at the corresponding time [4].

4.2. Nonuniqueness of the transporting velocity

For a prescribed density path p_t, the continuity equation determines $\operatorname{div}(p_t v_t)$. Several velocity fields may have the same weighted divergence [4, 16]. Indeed, if v_t satisfies the continuity equation and w_t satisfies

$$\operatorname{div}(p_t w_t) = 0,$$

then

$$\partial_t p_t + \operatorname{div}(p_t(v_t + w_t)) = \partial_t p_t + \operatorname{div}(p_t v_t) + \operatorname{div}(p_t w_t) = 0.$$

Conversely, the difference of two velocity fields representing the same density variation has zero weighted divergence. Among admissible fields, the WO construction selects a representative of minimum kinetic energy [4, 16].

4.3. The Wasserstein–Otto metric and path energy

The Wasserstein–Otto metric acts on these tangent vectors. Using the continuity equation, its squared norm along a smooth density path is [4, 16]

$$\boxed{\|\partial_t p_t\|^2_{p_t,\mathrm{WO}} = \inf_{v:\,\operatorname{div}(p_t v) = -\partial_t p_t} \int_E \|v(y)\|^2 p_t(y)\,dy.} \tag{11}$$

Weighted Helmholtz decomposition. For vector fields in $L^2(p; E)$, consider the scalar product

$$\langle u, v\rangle_{L^2(p)} = \int_E u(x) \cdot v(x)\, p(x)\,dx.$$

The weighted Helmholtz decomposition splits a field orthogonally into

$$v = \nabla\phi + w, \qquad \operatorname{div}(pw) = 0, \qquad \langle \nabla\phi, w\rangle_{L^2(p)} = 0;$$

see [17, Section 6]. In the general $L^2(p)$ setting, the gradient component belongs to the closure of smooth compactly supported gradients, and the divergence condition is distributional. We use $\nabla\phi$ for this component in the smooth setting. Consequently,

$$\int_E \|v\|^2 p\,dx = \int_E \|\nabla\phi\|^2 p\,dx + \int_E \|w\|^2 p\,dx.$$

Applied at each time with $p = p_t$, this decomposition identifies the minimiser in (11): the gradient component represents the same density variation and has minimum kinetic energy [17].

For a full path $t \mapsto p_t$, $0 \le t \le 1$, its energy is [16]

$$\mathcal{E}_{\mathrm{WO}}(p) = \int_0^1 \|\partial_t p_t\|^2_{p_t,\mathrm{WO}}\, dt = \int_0^1 \inf_{v:\, \mathrm{div}(p_t v) = -\partial_t p_t} \int_E \|v(y)\|^2 p_t(y)\, dy\, dt. \tag{12}$$

4.4. Minimum-energy velocity for an isotropic Gaussian initial law

Let $X \sim p_{\mathrm{data}} \in \mathcal{P}_+(E)$ have finite second moment, and let $Z \sim p_{\mathrm{init}} = \mathcal{N}(0, \mathrm{Id})$ be independent of X. No Gaussian assumption is made on p_{data}. For $X_t = (1-t)X + tZ$ and $0 < t < 1$, Gaussian convolution gives a smooth positive density. Differentiating its integral expression gives

$$\nabla p_t(y) = -\frac{1}{t^2} \int_E (y - (1-t)x) p_{\mathrm{data}}(x) \frac{1}{t^d} p_{\mathrm{init}}\left(\frac{y-(1-t)x}{t}\right) dx,$$

and hence

$$\nabla \log p_t(y) = -\frac{1}{t}\mathbb{E}[Z \mid X_t = y].$$

This Gaussian score relation also appears in stochastic interpolants [3]. Combining it with $X_t = (1-t)X + tZ$ and $v_t(y) = \mathbb{E}[Z - X \mid X_t = y]$ yields

$$v_t(y) = -\frac{y + t\nabla \log p_t(y)}{1-t} = \nabla \phi_t(y), \qquad \phi_t(y) = -\frac{\|y\|^2}{2(1-t)} - \frac{t}{1-t} \log p_t(y).$$

The field has finite $L^2(p_t)$ norm under the second-moment assumption. It is therefore the gradient representative in the weighted Helmholtz decomposition and minimises the kinetic energy among fields representing $\partial_t p_t$ [17]. This holds for every such p_{data}, including the Gaussian mixture used in our figures.

4.5. Geodesics in Wasserstein–Otto geometry

A constant-speed geodesic between p_{data} and p_{init} minimises the kinetic action. For endpoints with finite second moments, the Benamou–Brenier formula gives [5]

$$W_2^2(p_{\mathrm{data}}, p_{\mathrm{init}}) = \inf_{\substack{\partial_t p_t + \mathrm{div}(p_t v_t) = 0 \\ p_0 = p_{\mathrm{data}},\ p_1 = p_{\mathrm{init}}}} \int_0^1 \int_E \|v_t(y)\|^2 p_t(y)\, dy\, dt, \tag{13}$$

where the admissible pairs satisfy the continuity equation weakly and have finite action. If the geodesic density is smooth and positive and its minimum-energy velocity $v_t^{\mathrm{geo}} = \nabla \phi_t$ is continuously differentiable in time and space, this velocity satisfies [17]

$$\boxed{\partial_t v_t^{\mathrm{geo}} + (D v_t^{\mathrm{geo}}) v_t^{\mathrm{geo}} = 0.} \tag{14}$$

Here $D v_t^{\mathrm{geo}}$ is the spatial Jacobian matrix.

Figure 5 compares the independent Flow Matching path with displacement interpolation along a numerical optimal coupling of the same mixture and Gaussian endpoints [6]. We approximate each endpoint by 16384 equally weighted samples and minimise their squared-distance assignment cost. The

assigned particles move along straight segments; Gaussian smoothing of their mass and flux, with bandwidth 0.16, gives the intermediate densities and velocities. At $t = 0$ and $t = 1$, the known endpoint densities are displayed exactly. This approximates the geodesic between the continuous endpoint distributions. For the particle comparison, we select the same 150 Gaussian points from the numerical coupling. We integrate the reversed Flow Matching field with a fourth-order Runge–Kutta method and follow the assigned straight segments for optimal transport.

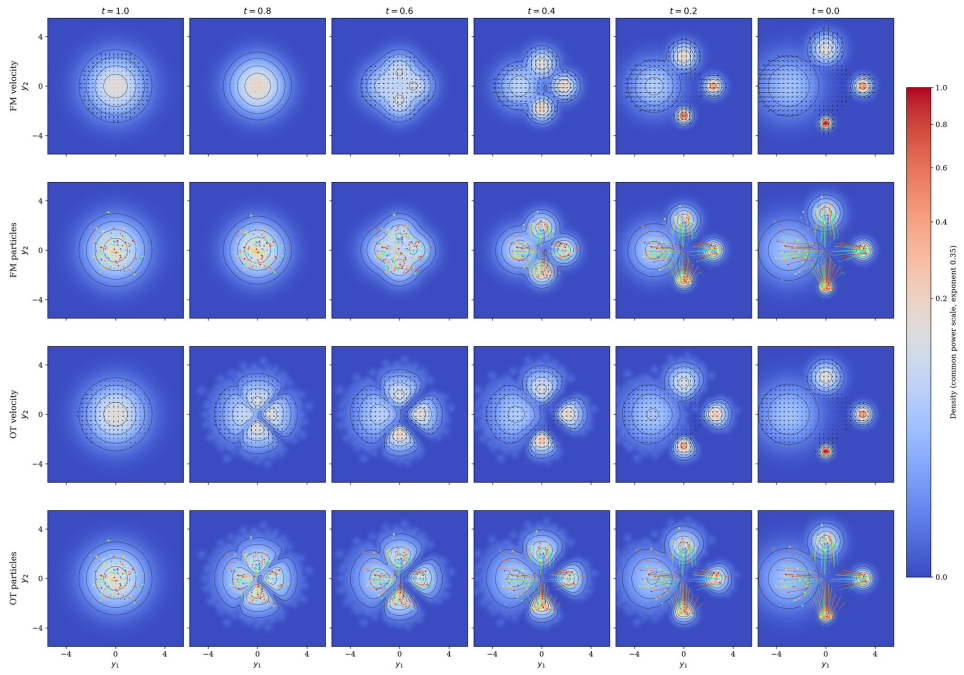


FIGURE 5. Flow Matching velocity and particle trajectories (rows 1–2), and numerical WO velocity and particle trajectories (rows 3–4). Both methods start from the same 150 Gaussian points, with matching particle colours. Time runs from $t = 1$ (left) to $t = 0$ (right); arrows show the reverse-time velocity and curves retain past trajectories.

5. Discussion

5.1. Minimum-energy velocity and a nongeodesic path

For $p_{\text{init}} = \mathcal{N}(0, \mathrm{Id})$, the exact population Flow Matching field is a gradient and is therefore the minimum-energy representative of each density variation, as shown in Section 4.4. A learned network approximates this field. The density path can nevertheless have excess action. Take independent $X, Z \sim \mathcal{N}(0, \mathrm{Id})$. Our interpolation formula gives

$$p_t = \mathcal{N}\Big(0, \big((1-t)^2 + t^2\big)\,\mathrm{Id}\Big), \qquad v_t(y) = \frac{2t-1}{(1-t)^2 + t^2}\, y.$$

The Gaussian contracts up to $t = 1/2$ and then expands. Its WO energy is positive, whereas the geodesic between these identical endpoints is constant, with zero energy. Figure 6 illustrates the difference.

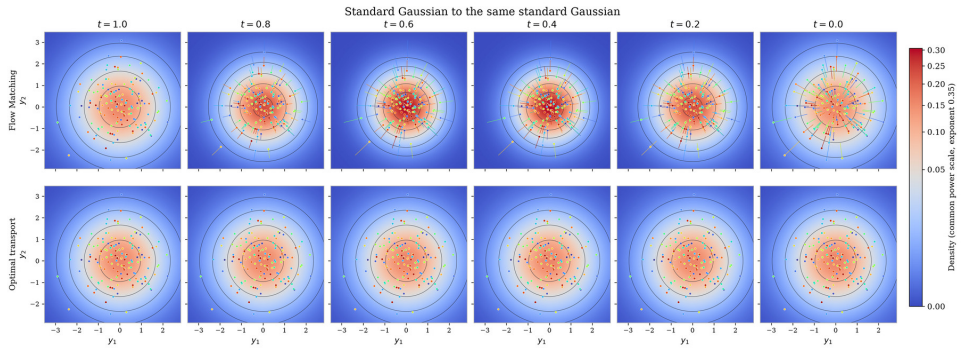


FIGURE 6. Transport between identical standard Gaussians. Flow Matching (top) contracts and expands; optimal transport (bottom) leaves particles fixed. Both rows use the same initial points; time runs from $t = 1$ to $t = 0$.

5.2. Straight trajectories and training towards optimal transport

For quadratic optimal transport in Euclidean space, particles along a Wasserstein geodesic follow affine trajectories $Y_t = (1 - t)Y_0 + tY_1$ [6]. For a smooth flow $\dot{Y}_t = v_\theta(t, Y_t)$, this means

$$\ddot{Y}_t = \big[\partial_t v_\theta + (Dv_\theta)v_\theta\big](t, Y_t) = 0.$$

A possible training regularisation is therefore

$$\mathcal{L}_{\mathrm{FM}}(\theta) + \lambda \int_0^1 \mathbb{E}\Big[\|\partial_t v_\theta(t, Y_t) + (Dv_\theta(t, Y_t))v_\theta(t, Y_t)\|^2\Big]\, dt, \qquad \lambda > 0,$$

where the expectation is over trajectories of the learned flow. This encourages constant velocities along particles. In several dimensions, straightness alone does not select the quadratic optimal coupling. To target OT, endpoint matching and the gradient structure of the geodesic velocity must also be controlled [17].

The independent-pair loss prescribes a generally nongeodesic path, so the added constraint changes its population minimiser. Using an OT coupling for the training pairs makes the regression target compatible with displacement interpolation [22]. These choices matter both for transport cost and for numerical sampling: an exactly constant velocity along a trajectory allows a single explicit Euler step to reach its endpoint, which motivates learning straighter flows [7].

6. Additional visualisations

We add two visualisations comparing Flow Matching and optimal transport, from a Gaussian initial law to a Laplace law and to a Gaussian mixture.

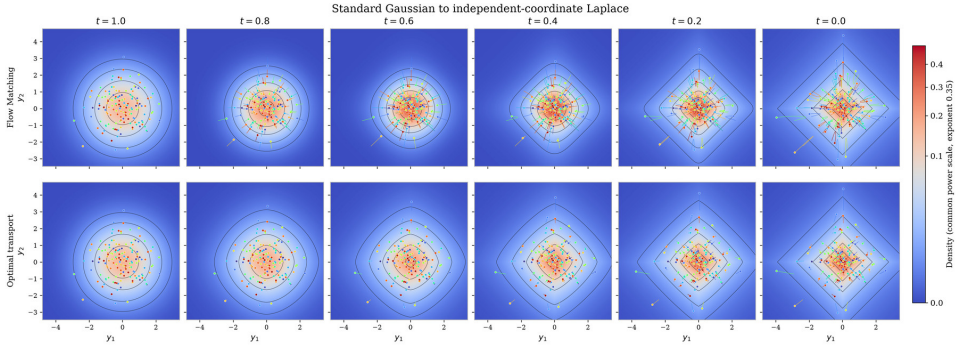


FIGURE 7. Gaussian to Laplace transport: Flow Matching (top) and optimal transport (bottom), with the same initial particles and past trajectories. Time decreases from $t = 1$ to $t = 0$.

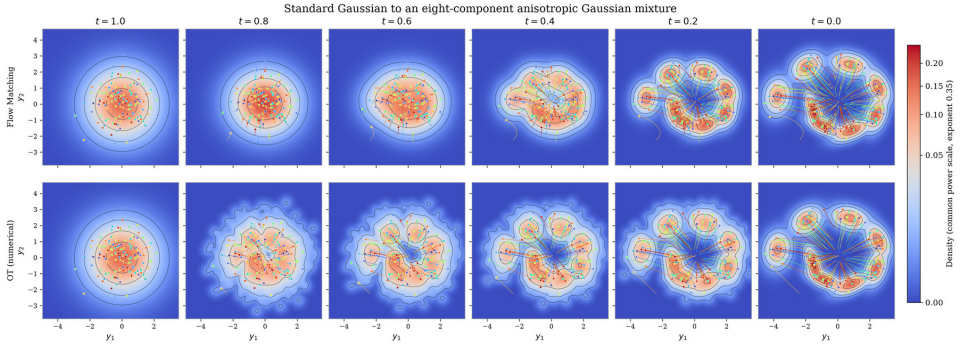


FIGURE 8. Gaussian to Gaussian mixture transport: Flow Matching (top) and numerical optimal transport (bottom), with the same initial particles and past trajectories. Time decreases from $t = 1$ to $t = 0$.

We also compare four initial laws transported to the same Gaussian mixture: a Gaussian, an independent-coordinate Laplace law, a uniform disk and a uniform square. All four are centred with covariance Id before numerical regularisation.

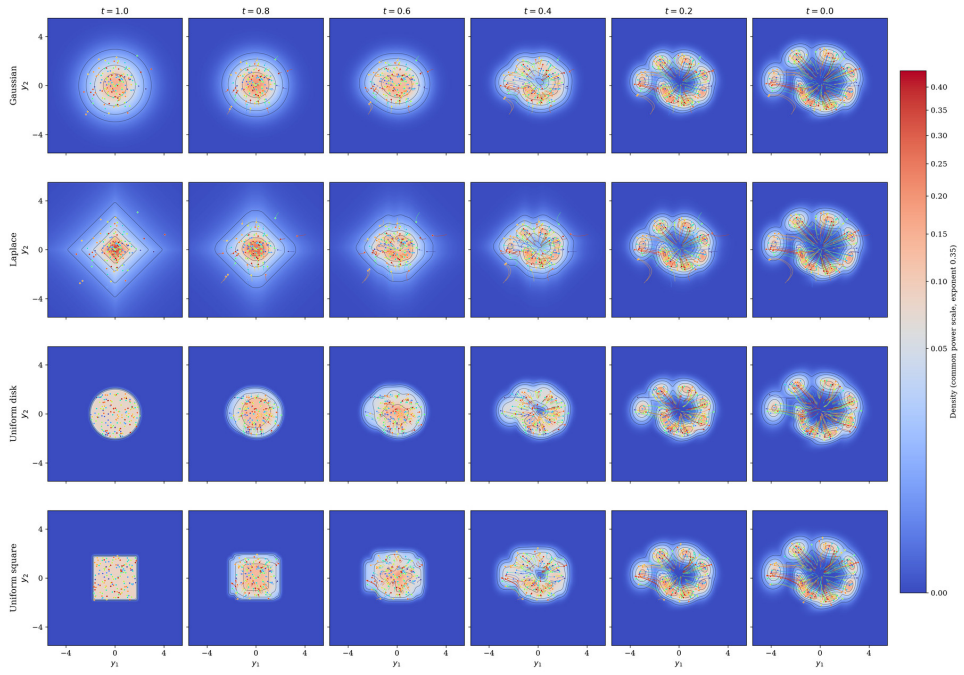


FIGURE 9. Flow Matching from four initial laws to the same Gaussian mixture. Rows show the Gaussian, Laplace, uniform disk and uniform square initial laws. Colours and contours show density; particles retain their past trajectories. Time decreases from $t = 1$ to $t = 0$. Endpoint laws are mollified with a Gaussian of standard deviation 0.06 for numerical evaluation.

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